KRISHNA V PREM

Electronics and Communication Engineer
• Indiranagar, Bengaluru, Karnataka, 560038 • +91 8078889016
• krishnavpremkumar@gmail.com • Portfolio • LinkedIn • GitHub • Leetcode

PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with strong proficiency in C/C++ and hands-on experience in Linux systems programming. Developed low-level applications including a TFTP client-server, Unix-like shell, and lexical analyzer, implementing process control, signal handling, and dynamic memory management. Currently undergoing advanced embedded systems training with exposure to microcontroller-based development.

SKILLS

- Programming Languages: C, C++
- Operating System: Linux
- Microcontroller: ESP32
- Networking: Socket Programming (UDP)
- Tools and Environment: VS Code, Git, MATLAB
- Data Structures and Algorithms: Linked lists, Hash tables, Tree, Search and sort algorithms
- PCB Design tools: Altium, EasyEDA, KiCAD

EDUCATION

Bachelor of Technology (Electronics and Communication with minor in ML) *2021 - 2025*
Amal Jyothi College of Engineering (Autonomous), Kottayam, Kerala **CGPA 8.07/10**
(Affiliated to APJ Abdul Kalam Technological University)

12th grade - Central Board of Secondary Education *2020 – 2021*
De Paul Public School Rajamudi, Idukki, Kerala **86%**

10th grade - Central Board of Secondary Education *2018 – 2019*
De Paul Public School Rajamudi, Idukki, Kerala **96%**

PROFESSIONAL TRAINING

Advanced Embedded Systems Training *June 2025 – Present*
Emertxe Information Technologies, Bengaluru, Karnataka

- Completed modules in Linux systems programming, C, C++, and Data Structures, currently advancing into PIC microcontroller programming and embedded hardware interfacing.

PROJECTS

Trivial File Transfer Protocol *February 2026*
- Developed a TFTP client-server application in C on Linux using POSIX UDP sockets, implementing RRQ/WRQ request handling, 512-byte block-based data transfer, sequential ACK with timeout/retry logic, and error packet processing for file-not-found, access violations, and disk-full conditions.

Mini Shell *January 2026*
- Developed a Unix-like shell in C with support for built-in and external command execution, input/output redirection, multi-stage pipelines, environment variable handling, and SIGINT/SIGCHLD signal management using fork, execvp, dup2, pipe, and waitpid for robust process control and terminal fidelity.

Comprehensive Banking System *January 2026*
- Developed a console-based banking system in C++ with a modular class hierarchy featuring an abstract Account base class extended by Savings Account, separate Customer and Banker roles with password authentication, timestamped transaction logging, custom exception classes for insufficient funds and authentication failures, and file I/O for persistent data storage across sessions.

Text Editor *January 2026*

- Developed a console-based text editor in C using a doubly linked list to store and manipulate text lines, supporting cursor navigation, insert/delete operations, global search and replace, and file open/save functionality, with a undo/redo system implemented using dynamic array stacks to track and reverse insert and delete actions.

Inverted Search Engine *December 2025*

- Developed an inverted search engine in C using hash tables with chaining and linked lists to index keywords across multiple documents, implementing word normalization, per-document frequency tracking, dynamic memory management, and efficient collision resolution for fast and scalable keyword retrieval.

Lexical Analyzer *December 2025*

- Developed a lexical analyzer in C that reads a C source file character by character, tokenizing and classifying lexemes into keywords, identifiers, operators, constants, and special characters, with support for all numeric literal formats (decimal, hexadecimal, octal, binary, floating-point with scientific notation), single and multi-line comment handling, string and character literal parsing, and detailed error reporting for malformed tokens.

MP3 Tag Reader *November 2025*

- Developed a binary metadata parser in C for ID3v1 and ID3v2 tag formats, implementing byte-precise field extraction using fread and fseek, structured parsing of title, artist, album, and track fields, and resilient handling of corrupted frames and non-standard encodings across diverse MP3 files.

LSB Image Steganography *October 2025*

- Developed a steganography tool in C that conceals and retrieves secret text within BMP images by manipulating pixel-level LSBs, incorporating magic-string based authentication to verify payload integrity during decoding, with a clean command-line interface for encode and decode operations.

ACADEMIC PROJECTS

Automated Traffic Alert System (ATAS) *Jan 2024 – June 2024*

- Developed an ESP32-based system where an emergency vehicle broadcasts an alert wirelessly to nearby vehicles, triggering real-time buzzer notifications to prompt drivers to clear the path, with firmware developed in Arduino IDE interfacing GPIO and wireless communication modules.

INTERNSHIP AND WORKSHOPS

- **Internship** (2023): ReverTech IT Solutions-PCB Designing
- **Workshop** (2024): 3D Printing: Design to Hardware (Hands-on), Dept. of ECE, AJCE

ACHIEVEMENTS

- Served as an Electrical Team member of Team Stellar, representing AJCE in the Solar Electric Vehicle Championship (SEVC) organized by the Coimbatore Society of Racing Minds (CSRM).